



# › PROJECT 1: NAIVE BAYES CLASSIFIER

Praktikum Maschinelles Lernen

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# PROJECT 1: NAIVE BAYES CLASSIFIER

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1. (5 Points) Implement a Python class NaiveBayes.

- > The class contains an `__init__`, `fit`, `predict_probability`, and `evaluate_on_data` method.
  - > The implementation should be able to deal with, both, discrete and continuous features. A template file with correspondent signatures and documentation will be handed out.
  - > Design your model generically: It should be able to deal with any number of classes and any number feature values (even though the data set contains only two classes and boolean discrete features).
  - > Don't change the given function signatures. Feel free to implement as many other methods as you need.
  - > **The only packages you're allowed to import for this task is Pandas and Math**
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2. (2 Points) Evaluate your Naive Bayes implementation using the „inflammation\_diagnosis.csv“ data set.
- > Combine the columns „inflammation“ and „nephritis“ to a new column „deasease“ which is „False“ if „inflammation“ and „nephritis“ is „False“ and „True“ otherwise. Use the column „deasease“ as target for the classifer and drop the original columns „inflammation“ and „nephritis“.
  - > Split the data into a train and a test set using 20% of the data for the test set. Show that no train data samples are contained in the test set.
  - > Fit the Naive Bayes classifier using the train set and evaluate it with the test set.
  - > Calculate and plot the confusion matrix.
  - > **You're only allowed to import your NaiveBayes class, Pandas, and Matplotlib (or Seaborn) for this task**
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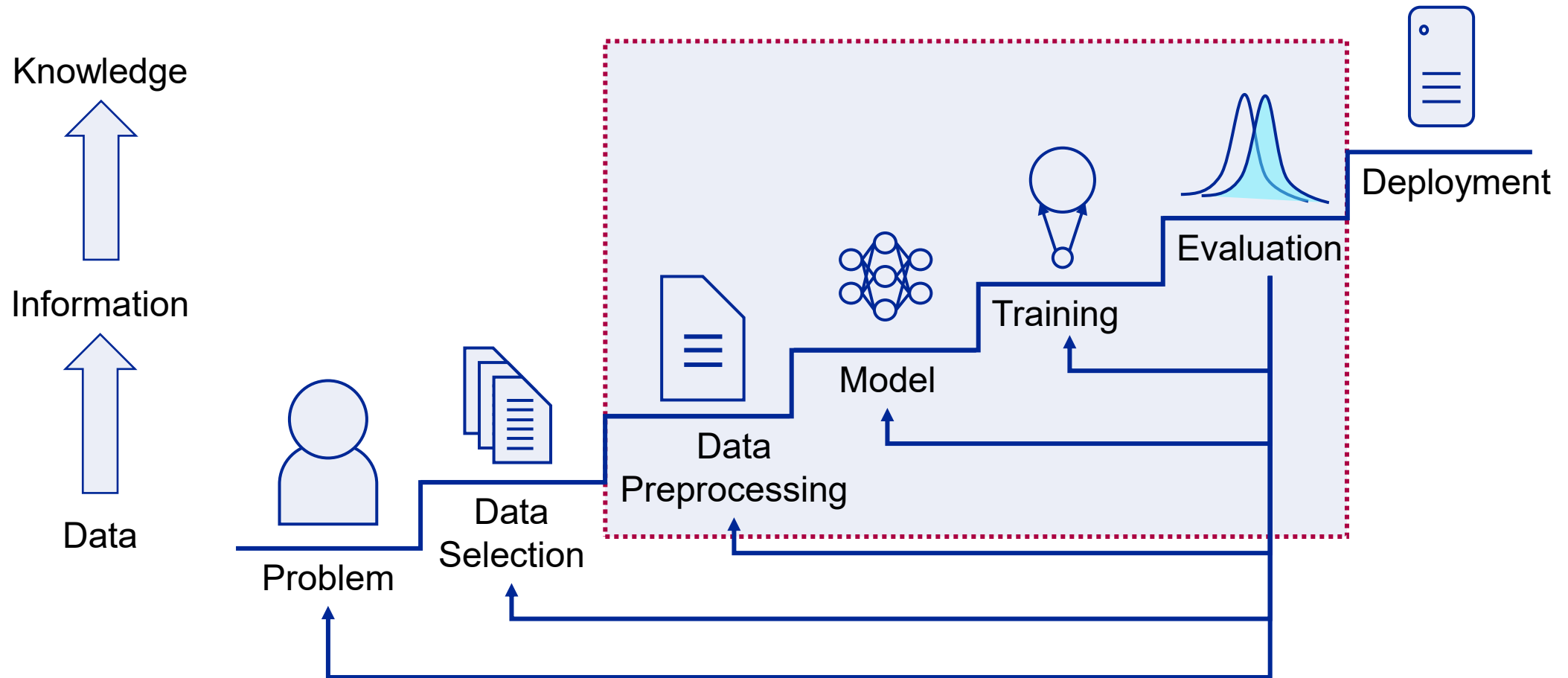
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- The submission consists of two files:
    1. A file „naiver\_bayes\_klassifizierer.py“ containing the NaiveBayes class implementation (task 1)
    2. A Jupyter Notebook containing the preprocessing, the training, and the evaluation of the Naive Bayes classifier (task 2)
  - Due Date is **October 24th**
  - **Remember: There are 3 Points for presentation, style, and creativity**
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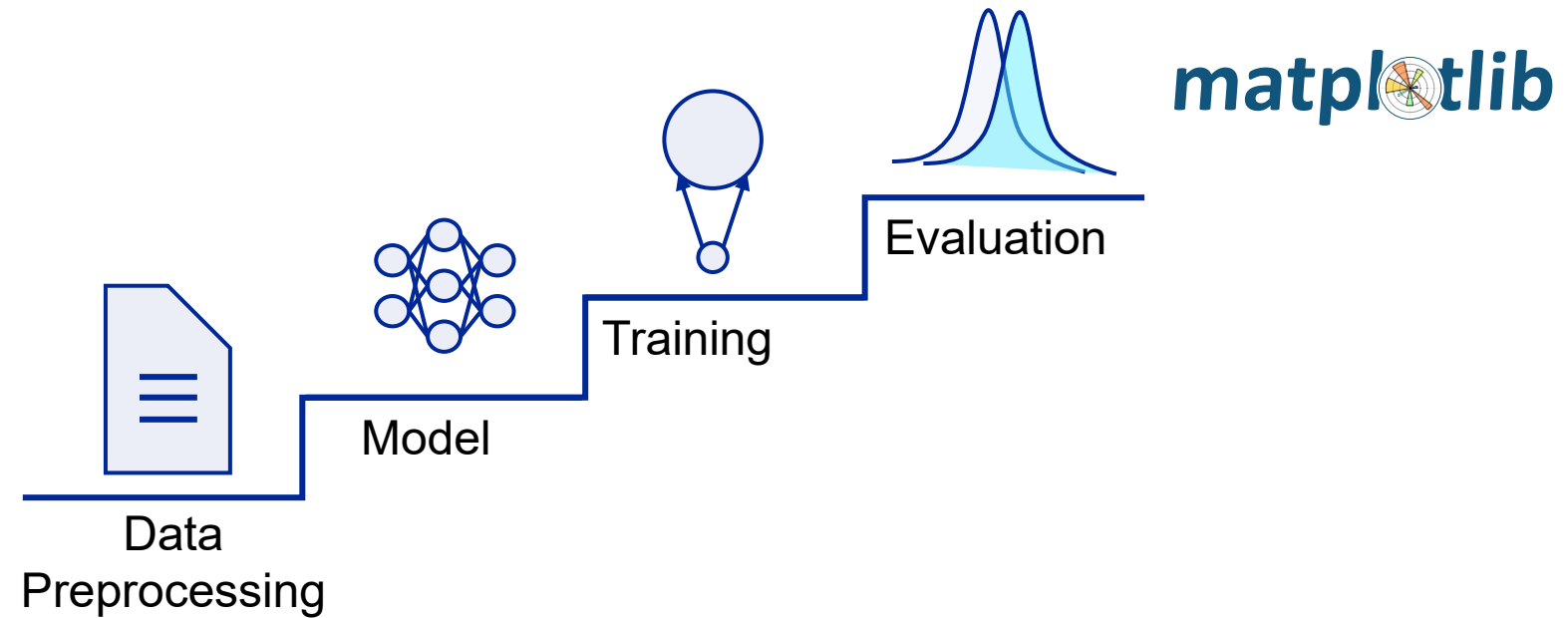
# WHAT'S TO BE LEARNED

## THE MACHINE LEARNING PROCESS



# THE TOOLS

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# WHAT'S TO BE LEARNED

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- **Learn how to extract information from data**
    - Deal with different data types (structured and unstructured)
    - ML algorithms
    - Explore
    - Statistics
    - Linear algebra (why live in 3 dimensions, when you can master 1000s?)
    - Use analysis to find the sweet spots
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# WHAT'S TO BE LEARNED

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- **Learn how to interpret your findings**
    - What are the methods and metrics?
    - What correlations can be found?
    - What's significant and what's not?
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# WHAT'S TO BE LEARNED

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- **Communicate and present your results**
    - Make pretty graphs
    - Your data contains a story. Tell it!
    - Make mistakes, harness criticism, defend your theses
    - Convince your peers, boss, customer, client, referee ...
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